

Mutable Variables

CS 350

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Curly-Mutvar

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- Learning goals

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 - To understand the design choices around functions in such a language
- Key concepts
 - Pass-by-reference vs. Pass-by-value

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- To allow variables values to change, we can make one simple change:
 - **Keep locations instead of values in the environment**
 - Then each variable refers to a single store location, whose value can change

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- x starts out as 3
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- End result is 6, the new value for x

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```
(define (env-lookup [n : Symbol] [env : Env]) : Location ....)
```

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- Static vs. Dynamic
 - What variables are in scope is a *static* property
 - Will always be the same for each call to a given function
 - What values are in memory is a *dynamic* property
 - Depends on every bit of code that has run before

Interpreting Variables

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(define (interp [env : Env]
                [e : Exp]
                [sto : Store]) : Result
  (type-case Exp e
    [(varE x)
     (v*s (lookup (lookup x env) sto) sto)]
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- Previously, we just looked up a value with `lookup`

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 - Recall, `lookup` is polymorphic
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 - So `lookup` on `env` gives `Location`, `lookup` on `store` gives `Value`
- Produce that value, along with the unchanged store

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(define-type Exp
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- Interpret the new value for the variable
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- Lookup up the location for the variable
- Return a store with the new value at that location

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- *What if the argument is already a variable?*

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 - Environment now takes *locations*
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 - Need to make sure the argument value ends up at that location
- If the argument is e.g. a number, then we have to make a new location for it
- *What if the argument is already a variable?*
 - Have a design decision

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 - If the arguments are variables, their values are looked up and copied to the new location
 - e.g. the normal way to interpret a variable
- Each function call has:
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 - Pointing to a new location

Pass-by-value interp

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[(appE funExpr argExpr)
  (with ([fun-v fun-sto] (interp env funExpr sto))
    ([arg-v arg-sto] (interp env argExpr fun-sto))
    (type-case Value fun-v
      [(closureV argVar funBody funEnv)
        (let* ([argLoc (new-loc arg-sto)]
               [body-sto (override-store argLoc arg-v arg-sto)]
               (interp (extend argVar argLoc funEnv)
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          [else error 'interp "Tried to call non-function"])]))]
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- Evaluate function and arg
 - Gives value and stores, threaded linearly

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- New location argLoc generated for function parameter

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- Evaluate function and arg
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- Evaluate function body

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 - Gives value and stores, threaded linearly
- New location `argLoc` generated for function parameter
 - New store `body-sto` with Parameter value added to this location
- Evaluate function body
 - In env from the closure
 - Extended with new location for var
 - With the store containing `arg-v` at location `argLoc`

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 - function's body is evaluated in an environment *where the argument variable is bound to the input variable's location*
- Changes to one will be seen in the other

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;; Everything the same as pass-by-value  
;; except we check if the argument is a variable  
;; and use its location instead of the new one if it is  
(type-case Exp argExpr  
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 - Evaluate function body in closure environment
 - Extended to have parameter symbol mapped to **the variable's existing location**

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 - e.g. Write a function that says “change these variables in this way” that can be used over and over again
 - e.g. swap two variable’s values

Example

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```
{let1 {y 2}
  {let1 {f {lam x {seq
                    {set-var! x {* x 3}}
                    x}}}}
    {+ {f y} y}}}
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- The final result is 6, since the value of y was changed

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 - Its location contains 100
 - NOT used in the function call, because not in scope for function definition

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- These properties are **completely separate from each other**
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 - But lazy + mutable state is generally a bad idea